Tethered C6 to Satellite Wiring Diagram

This diagram shows how to wire a tethered/usb-sniffer primary C6 to one or more satellite C6 boards over SPI.
Pin mapping matches firmware:

- firmware/src/standalone/spi_master.h (primary pin map used by usb-sniffer OTA relay path)
- firmware/src/satellite/transport_spi.h (satellite pin map)

1) Single Satellite (minimum wiring)

[Mermaid Wiring Diagram]

```
P6 --> S16
P7 --> S17
S12 --> P2
P10 --> S110
S13 --> P20
P23 --> S111
P5V --- S15V
PGND --- S1GND
```

2) Up to 3 Satellites (full carrier-style wiring)

Shared across all satellites:

- GPIO6 SCK
- GPIO7 MOSI
- GPIO2 MISO
- GPIO23 SYNC
- 5V and GND

Per-satellite control lines:

- CS: GPIO10 (sat 1), GPIO18 (sat 2), GPIO19 (sat 3)
- DATA_READY: GPIO20 (sat 1), GPIO21 (sat 2), GPIO22 (sat 3)

[Mermaid Wiring Diagram]

```
PSCK --> ASCK
PSCK --> BSCK
PSCK --> CSCK
PMOSI --> AMOSI
PMOSI --> BMOSI
PMOSI --> CMOSI
AMISO --> PMISO
BMISO --> PMISO
CMISO --> PMISO
PSYNC --> ASY
PSYNC --> BSY
PSYNC --> CSY
PCS1 --> ACS
PCS2 --> BCS
PCS3 --> CCS
ADR --> PDR1
BDR --> PDR2
CDR --> PDR3
P5V --- A5V
P5V --- B5V
P5V --- C5V
PG --- AG
```

```
PG --- BG
PG --- CG
```

Connection Table (all necessary nets)

Net	Primary (tethered C6)	Satellite pin	Required	Direction
SCK	GPIO6	GPIO6	Yes	Primary -> Satellite
MOSI	GPIO7	GPIO7	Yes	Primary -> Satellite
MISO	GPIO2	GPIO2	Yes	Satellite -> Primary
CS1	GPIO10	GPIO4 (sat 1)	Yes (sat 1)	Primary -> Satellite
CS2	GPIO18	GPIO4 (sat 2)	Optional (sat 2)	Primary -> Satellite
CS3	GPIO19	GPIO4 (sat 3)	Optional (sat 3)	Primary -> Satellite
DREADY1	GPIO20	GPIO3 (sat 1)	Yes (sat 1)	Satellite -> Primary
DREADY2	GPIO21	GPIO3 (sat 2)	Optional (sat 2)	Satellite -> Primary
DREADY3	GPIO22	GPIO3 (sat 3)	Optional (sat 3)	Satellite -> Primary
SYNC	GPIO23	GPIO5	Yes	Primary -> Satellite
Power	5V	5V	Yes	Power rail
Ground	GND	GND	Yes	Common reference

Wiring Notes

- Use a common ground for every board.
- Keep SPI wires short and paired with nearby ground where possible.
- Shared MISO is safe because only the selected satellite drives it while its CS is active.
- Avoid using C6 strapping pins for custom remaps unless you also update firmware pin defines.
- Satellite radio_id values should be unique (1..3) when using multiple satellites.
- **The satellite's own CS/SYNC pins were originally GPIO10/GPIO11 -- verify against your board before wiring.** Common "ESP32-C6 SuperMini" boards (e.g. the MakerGO variant) only break out GPIO0-9 and GPIO12-23; GPIO10 and GPIO11 have no pad at all. The satellite firmware now uses GPIO4 (CS) and GPIO5 (SYNC) instead -- both confirmed broken out. If your specific SuperMini variant differs, cross-check its silkscreen/pinout before wiring, and update firmware/src/satellite/transport_spi.h to match.